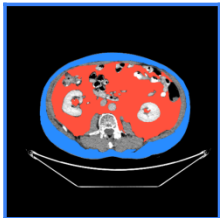


Systematic study of data-scarce body-composition segmentation

TWO PRIMARY DATASETS

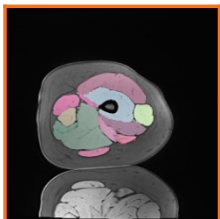
CT + SAT/VAT



AATTCT-IDS (CT)

300 subjects · 3213 slices
512×512 · SAT/VAT (3 cls)

MRI + 13 muscles



Thigh Dixon MRI

25 subjects · 1513 slices
256×256 · 4ch · 14 cls

FOUR EXPERIMENTAL FACTORS

Factor 1 . Architecture

U-Net | DeepLabV3+ | HRNet-W18

Factor 2 . Pretraining source

random | ImageNet | CT
self-supervised (SSL)

Factor 3 . Input design

+FF (primary) · +SE gate (ablation)

Factor 4 . Multi-task learning

shared encoder,
two task decoders

EVALUATION

Fixed held-out benchmark

architecture, 2.5D,
shared encoder

Repeated five-fold CV

pretraining,
input design

Statistical unit: subject

3 seeds -> mean ± SD